

“Discrete survival analysis”

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Discrete survival analysis

Suppose that event times can only happen at times $\tau_1, \dots, \tau_K, \infty$, where ∞ means that no event was observed within $[0, \tau_K]$. Events can only occur at each time point τ_j . The survival function at time k in this context is defined analogously to the continuous case:

$$S(k) = \Pr(T > k).$$

However, the discrete-time hazard function at time k is a conditional probability.

$$h(k) = \Pr(T = k \mid T \geq k).$$

Some useful relationships are presented below (which are derived from the above definition and the definition of conditional probability):

$$\begin{aligned}\Pr(T = k) &= h(k)S(k-1), \\ S(k) &= S(k-1)(1 - h(k)), \\ S(k) &= \prod_{j=1}^k (1 - h(j)).\end{aligned}$$

Taking logarithms on both sides we get

$$\log S(k) = \sum_{j=1}^k \log(1 - h(j)).$$

This leads to the expression of the cumulative hazard:

$$H(k) = -\log S(k) = -\sum_{j=1}^k \log(1 - h(j)).$$

Pooled logistic model

Consider the model for the hazard (probability of receiving treatment)

$$\begin{aligned} \text{logit } h(k \mid \mathbf{X}_{ik}) &= \text{logit } \Pr(T = k \mid T \geq k, \mathbf{X}_{ik}) \\ &= \alpha_k + \mathbf{X}_{ik}, \end{aligned}$$

where α_k are the time-specific intercepts (baseline hazard), and $\mathbf{X}_{ik}$ are the covariates.

Sequential Propensity Scores as Discrete-Time Hazards

Setup

- Let T be the time of treatment initiation: $T \in \{1, \dots, K, \infty\}$.
- Define $A_k = 1$ if the individual is treated at time (window) k , 0 otherwise.
- Let $\bar{A}_{k-1} = 0$ indicate not treated before month k .
- Let $\bar{L}_k$ be the covariate history up to month k (if available, otherwise we may need to use baseline covariate values).

By construction:

$$A_k = 1 \iff T = k$$

Discrete-Time Hazard Function

The **discrete-time hazard of treatment** at time-window k is:

$$h(k \mid \bar{L}_k) = \Pr(T = k \mid T \geq k, \bar{L}_k)$$

Interpretation: probability of receiving treatment in time-window k , conditional on not having been treated yet.

Sequential Propensity Score

The **sequential propensity score** is defined as:

$$e_k(\bar{L}_k) = \Pr(A_k = 1 \mid \bar{A}_{k-1} = 0, \bar{L}_k)$$

Notice that:

$$\{T \geq k\} \equiv \{\bar{A}_{k-1} = 0\}$$

Therefore:

$$\boxed{e_k(\bar{L}_k) = h(k \mid \bar{L}_k)}$$

Implication: estimating a sequential propensity score is equivalent to estimating the discrete-time hazard of treatment.

Connection to Weights

The probability of a specific treatment history up to month k can be factorized as:

$$\Pr(T = k \mid \bar{L}_k) = h(k \mid \bar{L}_k) \prod_{j < k} (1 - h(j \mid \bar{L}_j)).$$

To check: **inverse probability weights** in longitudinal causal inference.